MILITARY SPECIFICATION

DETECTOR-DEWAR DT-617/UA

This specification is approved for use by USAERADCOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of common module, Detector-Dewar DT-617/UA.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of
MIL-C-49175	- Cooler, Cryogenic, Mechanical HD-1033/UA
MIL-E-55585	- Electronic Equipment and Parts, Packaging of

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	- Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-726	- Packaging Requirement Codes
MIL-STD-781	- Reliability Tests: Exponential Distribution
MIL-STD-810	- Environmental Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, USA Electronics Research and Development Command, ATTN: DRDEL-E, 2800 Powder Mill Road, Adelphi, MD 20783 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5855

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

DRAWINGS

USAERADCOM

5002476	- Detector-Dewar DT-617/UA
DL-5002476	- Detector-Dewar DT-617/UA

2.2 Non-Government documents. The following document, of the exact issue shown, forms a part of this specification to the extent specified herein. In the event of conflict between the document referenced herein and the contents of this specification, the contents of this specification shall be considered a superseding requirement.

Standard Procedures for Testing Infrared Detectors and Their Performance, R.C. Jones, D. Goodwin, and G. Pullen, dated 12 September 1960, Office of the Director of Defense Research and Engineering.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The Detector-Dewar DT-617/UA, referred to herein as the detector-Dewar, is an opto-electronic device sensitive to radiation in the 7.5- to 12.0-micrometer spectral region. The detector-Dewar shall use a 120-element detector array of photoconductive mercury-cadmium-telluride operated from a supply of +5.0, ±0.2 volts direct current (Vdc). The detector bias resistors shall be an integral part of the detector-Dewar.

3.2 Construction. The detector-Dewar shall be constructed in accordance with DL-5002476 and as specified herein.

3.2.1 Weight. Weight of the detector-Dewar shall not be greater than 0.75 pound including bias resistors.

3.3 First article. When specified in the contract (see 6.2) the contractor shall furnish first article units for inspection (see 4.3).

3.4 Materials, parts, and processes. Materials, parts, and processes shall be as specified herein and as shown on the applicable drawings.

3.5 Components. The detector-Dewar shall consist of a 120-element detector array and a resistor bias pack. The detector array orientation with respect to the Dewar mounting flange shall be variable through 360 degrees. At least two reusable, electrically activated getters shall be an integral part of the detector-Dewar for periodic maintenance of the vacuum, and each getter shall withstand at least seven firings after shipment. The getter activation current (amperes) and time (minutes) shall be provided on the label specified in 5002476.

3.6 Performance characteristics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

3.6.1 Detector-Dewar defects. Unless otherwise specified herein, the detector-Dewar shall be operated at 80 (+5, -20) K. The detector-Dewar shall contain no more than 5 defective channels, as defined in 6.3. Defective channels shall be separated by at least 4 nondefective channels. Defective detector channels shall be electrically strapped to an adjacent channel with common ground. No defective or strapped channel shall be within the center 36 channels of the array.

3.6.2 Time constant. The time constant of each detector element shall not be greater than 2 microseconds.

3.6.3 Sensitive area. The sensitive area of the detector elements shall fall within the dimensions specified on 5002476.

3.6.4 Crosstalk. The crosstalk signal level between any two elements shall be less than -26 dB.

3.6.5 Detectivity. The 500 K blackbody detectivities shall be as follows:

a. At least 108 elements, not less than 8.5×10^{9} cm $Hz^{\frac{1}{2}} W^{-1}$.

b. The remaining 12 elements not less than 4.8×10^{9} cm $Hz^{\frac{1}{2}} W^{-1}$.

c. Average of all elements, not less than 1.1×10^{10} cm $Hz^{\frac{1}{2}} W^{-1}$.

3.6.5.1 Noise. The detector noise of each element shall be not less than 3.5×10^{-9} volts rms $Hz^{-\frac{1}{2}}$.

3.6.6 Detector resistance. The detector resistance of each element shall be not less than 20 ohms and not greater than 300 ohms.

3.6.7 Responsivity. The average of the 500 K blackbody responsivities of the acceptable elements shall be not less than 8.3×10^{3} volts/watt nor greater than 20×10^{3} volts/watt. The blackbody responsivity of each element shall not vary from the mean by more than ±40 percent of the mean value.

3.6.8 Temperature sensors. At least two calibrated 2N2222 temperature sensor chips, or equivalent, shall be installed at or near the detector array location in the Dewar to provide an electrical signal which is an analog of temperature. The temperature sensor shall have a junction voltage of 1.060, ±0.015 volt with a 1 mA ±10 percent forward current when the detector-Dewar is operated at 77, ±3 K.

3.6.9 Spectral response. Each element of the detector-Dewar shall have a spectral sensitivity of 7.7, ±0.25 to 11.75, ±0.25 micrometers.

3.6.10 Detector bias. The average bias current of the array shall not exceed 4.5 milliamperes per element, and no one element shall take a bias current of more than 8.0 milliamperes. The total bias power dissipated by the array shall not exceed 70 milliwatts. The detector resistor bias pack shall be operated at +5.0, ±0.2 Vdc. Each detector element shall be biased with current flow in the same direction.

3.6.11 Dewar warm and cold shields. The clear aperture (warm shield) shall be in accordance with the detail requirements of 5002476 and the following requirements: Cold shielding is optional (at the discretion of the detector manufacturer) but, when used, shall also conform to these same requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

a. In the direction of the array's axis (either (1) or (2)).

(1) For each element, the cold shield(s) shall allow no vignetting of any ray to the center point on the detector surface within an angle of 18 degrees measured from and on either side of a line perpendicular to the array's axis and parallel to the optical axis.

(2) Each element shall be allowed to accept radiation from a 1.00, ±0.05 inch diameter stop centered on the detector-Dewar's optical axis and located a maximum of 1.60 inches behind the plane of the detectors.

b. In the direction perpendicular to the array's axis.

The cold shield(s) shall allow for each element, no vignetting of any ray to any point on the detector surface within an angle of 23 degrees measured from and on either side of a line perpendicular to the array's axis and parallel to the optical axis.

3.6.12 Heat load. The detector heat load shall not exceed 0.37 watt when all 120 detector elements are operating at their final bias current and the Dewar is cooled to 80, ±5 K.

3.6.13 Vacuum integrity. The Dewar shall comply with heat load requirements of 3.6.12 for a period of at least one (1) year.

3.6.14 Microphonics. There shall be no peak in the 50 Hz - 20 kHz noise power spectrum in any 50 Hz bandwidth which exceeds 10 dB referenced to the white (flat) portion of the noise spectrum when the detector-Dewar is in full operating condition and cooled by a cryogenic cooler built in accordance with MIL-C-49175.

3.7 Environmental conditions.

3.7.1 Temperature shock. The detector-Dewar shall not be damaged (see 6.3.1) by sudden changes in temperature between -54°C and +71°C.

3.7.2 High temperature. The detector-Dewar shall not be damaged by storage or operation to +71°C.

3.7.3 Low temperature. The detector-Dewar shall not be damaged by operation to -54°C or storage to -62°C.

3.7.4 Shock. The detector-Dewar shall not be damaged by high intensity shocks of 100 g's peak amplitude with 11 milliseconds duration and by bench handling tests.

3.7.5 Vibration. The detector-Dewar shall not be damaged by vibration over the frequency spectrum at the specified g levels and amplitudes shown in figure 1.

3.8 Reliability. The detector-Dewar shall have a specified mean time between failure (MTBF) of at least 27,600 hours.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

3.9 Treatment and painting. Unless otherwise specified, the detector-Dewar shall be treated and painted in accordance with the applicable drawing.

3.10 Preconditioning. Each detector-Dewar shall be subjected to random vibration and temperature cycling prior to group A testing.

3.11 Nameplates and product marking. Unless otherwise specified, the detector-Dewar module, parts, components, subassemblies, and assemblies thereof shall be marked for identification and reference designation markings in accordance with the applicable drawing.

3.12 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.5).
c. Inspection of packaging (see 4.9).

4.3 First article inspection. When specified in the contract, the first article inspection shall be performed by the contractor on a minimum of 5 preconditioned (see 4.10) sample detector-Dewars.

4.3.1 Inspections. All materials, parts, processes and assemblies shall be examined for conformance to the applicable specification or drawing. Inspections shall be made using MIL-STD-105, General Inspection Level II with a 2.5 percent AQL for mechanical and 4 percent AQL for visual inspection, except as noted in table I.

TABLE I. Inspections.

Inspection Requirement	Percent AQL	Requirement paragraph	Inspection criteria
Any part or component missing or damaged	2.5	3.2, 3.4	MIL-STD-252
Weight not as specified	2.5	3.2.1	MIL-STD-252
Marking not as specified	2.5	3.11	
Workmanship not as specified	2.5	3.12	MIL-STD-454
Treatment and painting not as specified	2.5 1/	3.9	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

TABLE I. Inspections. - Continued

Inspection Requirement	Percent AQL	Requirement paragraph	Inspection criteria
Dimensions not as specified	2.5	3.2	
Getters and label not as specified	1.0	3.5	5002476
Detector orientation range not as specified	1.0	3.5	5002476
Sensitive area not as specified	2/	3.6.3	5002476
Dewar warm and cold shields not as specified	2/	3.6.11	5002476

1/ MIL-STD-105, Inspection Level S-3 or objective quality evidence.
2/ Inspect every array 100 percent and inspect at least 3 elements in each array.

4.3.2 Tests. First article inspection shall be conducted upon units which have successfully completed the inspections specified in 4.3.1. Prior to start of tests specified in table II, detector-Dewars shall be identified as to their applicable subgroups. A minimum of 2 detector-Dewars shall experience the testing of subgroup 1 and a minimum of 3 detector-Dewars shall experience the testing of subgroup 2. The testing of both subgroups may be performed concurrently. Tests may be conducted in any order, except that microphonics shall be performed after shock and vibration (4.7.4 and 4.7.5). Failure of any tests shall constitute first article failure.

TABLE II. First article inspection.

Inspection	Requirement paragraph	Test paragraph	Subgroup 1	Subgroup 2
Detector-Dewar defects	3.6.1	4.6.1	X	X
Time constant	3.6.2	4.6.2	X	X
Crosstalk	3.6.4	4.6.4	X	X
Detectivity	3.6.5	4.6.5	X	X
Detector resistance	3.6.6	4.6.6	X	X
Responsivity	3.6.7	4.6.7	X	X
Temperature sensors	3.6.8	4.6.8	X	X
Spectral response	3.6.9	4.6.9	X	X
Detector bias	3.6.10	4.6.10	X	X
Heat load	3.6.12	4.6.12	X	X
Vacuum integrity	3.6.13	4.6.13	X	X
Microphonics	3.6.14	4.6.14	X	X
Temperature shock	3.7.1	4.7.1	X	
High temperature	3.7.2	4.7.2	X	
Low temperature	3.7.3	4.7.3	X	
Shock	3.7.4	4.7.4	X	
Vibration	3.7.5	4.7.5	X	
Reliability	3.8	4.8		X

4.3.3 Disposition of first article samples. First article samples shall not be considered as part of the procurement quantities (see 6.2.h).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

4.4 Inspection procedures. Unless otherwise specified herein, the detector-Dewar shall meet the specified performance requirements at an ambient temperature of +23°C, ±5°C, when operated at a temperature of 80, ±5 K, and biased to +5.0, ±0.2 Vdc.

4.5 Quality conformance inspection.

4.5.1 Inspection. Inspection shall be as specified in 4.3.1.

4.5.2 Tests.

4.5.2.1 Group A inspection. Group A inspection shall be conducted on all detector-Dewars following successful inspection of 4.5.1 and preconditioning of 4.10. Group A tests listed in table III may be performed in any order except that detector-Dewar defects (4.6.1) shall be run last. Failure of any test shall be cause for rejection of that unit.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Microphonics	3.6.14	4.6.14
Detectivity	3.6.5	4.6.5
Detector resistance	3.6.6	4.6.6
Responsivity	3.6.7	4.6.7
Temperature sensors	3.6.8	4.6.8
Detector bias	3.6.10	4.6.10
Heat load	3.6.12	4.6.12
Detector-Dewar defects	3.6.1	4.6.1

4.5.2.2 Group B inspection. Group B inspection is not required.

4.5.2.3 Group C inspection. Group C inspection shall be conducted on detector-Dewars selected from units which have passed the tests in 4.5.2.1. The tests of 4.6.9 and 4.6.13 shall be performed on a sample quantity.

4.5.2.3.1 Sampling for group C inspection. The tests of 4.6.9 and 4.6.13 shall be initially performed at a reduced sample rate of 3 percent of the specified lot (see 6.3.8) or one unit, whichever is the larger quantity.

a. When zero failure occurs, the 3 percent or one unit inspection sampling shall continue.

b. Failure of any unit to pass the inspection of either 4.6.9 or 4.6.13 shall require 100 percent inspection of the entire lot for the failed attributes only with continued follow-on lots inspected until zero failure occurs in any lot. Upon achieving zero failure at 100 percent inspection, follow-on lot inspection shall be conducted at 10 percent sampling. Thereafter, upon achievement of zero failure during 10 percent sampling, the sample rate shall be lowered to 3 percent of the specified lot or one unit, whichever is larger. However, if one or more failures occur during the 10 percent inspection sampling, the 100 percent inspection testing shall be reinstituted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

4.5.2.3.2 Group C failures. Actions required relative to group C failures shall be as specified in the contract or purchase order (see 6.2).

4.5.2.3.3 Disposition of group C samples. Group C samples shall be accepted on contract subsequent to the tests of table III.

4.5.2.4 Group D inspection. Group D inspection not required.

4.6 Test methods.

4.6.1 Detector-Dewar defects. The detector array shall be checked for defects. Failure to meet requirements of 3.6.1 shall constitute failure of this test.

4.6.2 Time constant. Measurement shall be performed on 5 elements of the array each separated by at least 20 elements. The time constant is defined as the time required for the element response to reach 63 percent of its steady state value when irradiated by a 7.5 to 12 micrometer source which produces a signal-to-noise ratio not greater than 100:1. This time constant shall be considered equal to the time constant defined by the high frequency rolloff of the noise spectrum of the detector. Any measurement technique consistent with these definitions shall be considered acceptable. Failure to meet requirements of 3.6.2 shall constitute failure of this test.

4.6.3 Sensitive area. The physical area as defined by drawing 5002476 shall be determined by optical microscope examination. Measurement shall be made on 3 elements of the array each separated by at least 20 elements prior to the integration of the Dewar window with the detector-Dewar. Failure to meet requirements of 3.6.3 shall constitute failure of this test.

4.6.4 Crosstalk. Measurements shall be made on 4 elements of the array. The crosstalk shall be measured under the following conditions:

a. An element shall be irradiated by a 7.5 to 12 micrometer source having a signal level 20 times greater than the noise level for the element. The image of the source shall fall entirely within the area of that detector element.

b. The signal level of the element adjacent to and on the same ground path of the irradiated element shall be measured. The element under test shall meet the requirements of 3.6.4.

Failure to meet requirements of 3.6.4 shall constitute failure of this test.

4.6.5 Detectivity. Noise measurements shall be made on each element of the array under the following conditions: 295, $\pm$5 K background temperature and equivalent noise bandwidth no less than 34 kHz with a low frequency 3-dB point (high pass) no greater than 80 Hz. Using the results of these measurements, the results of the 500 K blackbody responsivity measurements, and the nominal detector element size (see 6.3.7), the 500 K blackbody detectivity for each element of the array shall be calculated. Failure to meet requirements of 3.6.5 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

4.6.6 Detector resistance. The resistance of each element of the array shall be measured. Failure to meet requirements of 3.6.6 shall constitute failure of this test.

4.6.7 Responsivity. Blackbody responsivity of each element of the array shall be measured. Nominal detector element size (see 6.3.7) shall be used in responsivity calculations and blackbody temperature shall be 500, ±5 K. Failure to meet requirements of 3.6.7 shall constitute failure of this test.

4.6.8 Temperature sensors. The 77 K calibration point shall be determined by cooling the inner Dewar with liquid nitrogen. One milliampere ±10 percent bias shall be applied to the sensor diode and the voltage shall be monitored and recorded at the 77 K calibration point. Failure to meet requirements of 3.6.8 shall constitute failure of this test.

4.6.9 Spectral response. The spectral response of 3 elements each separated by at least 20 elements shall be tested per "Standard Procedures for Testing Infrared Detectors and Describing Their Performance", by R. C. Jones, D. Goodwin, and G. Pullen, dated 12 September 1960, Office of the Director of Defense Research and Engineering. The spectral sensitivity is defined as the 50 percent relative response points and shall be measured on the assembled detector-Dewar. Failure to meet requirements of 3.6.9 shall constitute failure of this test.

4.6.10 Detector bias . The detector bias power shall be tested per "Standard Procedures for Testing Infrared Detectors and Describing Their Performance", by R. C. Jones, D. Goodwin, and G. Pullen, dated 12 September 1960, Office of the Director of Defense Research and Engineering. Failure to meet requirements of 3.6.10 shall constitute failure of this test.

4.6.11 Dewar warm and cold shields. During in-process inspection, the warm and cold shields shall be measured to ensure conformance with the requirement. Alignment of the cold shield relative to detector elements shall be checked 100 percent during in-process inspection. Failure to meet requirements of 3.6.11 shall constitute failure of this test.

4.6.12 Heat load. The heat load of the detector-Dewar shall be determined with all detectors operating at specified bias. The heat load of the detector-Dewar shall be determined by partially filling the Dewar's inner stem with liquid nitrogen and monitoring the rate of loss of the cryogen. The heat load shall be calculated from the average or integrated loss rate of the cryogen for the last 5 minutes of the Dewar's hold-time. For calculations, cooling produced by the nitrogen phase change only shall be considered as total useful energy. If this test is performed with the detector array unbiased, the results of this test shall be added to the results of the detector bias power test to determine the heat load. Failure to meet requirements of 3.6.12 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

4.6.13 Vacuum integrity. Heat load shall be measured on the unbiased detector-Dewar. The temperature of the detector-Dewar shall then be elevated to +71°C for a period of 72 hours. At the completion of the temperature exposure and stabilization at ambient temperature, the heat load shall be measured again. Getters may not be fired during this test. As it is not possible to test for long-term vacuum integrity, failure to meet the requirements of 3.6.12 within a period of one (1) year shall constitute a latent defect. Failure of the detector-Dewar to meet the requirements of 3.6.12 or a 15 percent increase in heat load between the measurement before and after temperature exposure shall constitute failure of this test.

4.6.14 Microphonics. The detector-Dewar shall be coupled to a mechanical cryogenic cooler (MIL-C-49175) and operated in the cooled condition of 60-80 K with the channels under test at full bias. All elements shall be checked using a noise power spectrum analyzer with 50, ±5 Hz noise bandwidth between 50 Hz and 20 kHz. Failure of one or more channels in the 12 channel sample shall be reason for rejection of the detector-Dewar. Failure to meet requirements of 3.6.14 shall constitute failure of this test.

4.7 Environmental tests. The environmental tests shall be performed in accordance with MIL-STD-810 as modified by this specification. The performance tests required by MIL-STD-810 before and after each environmental test shall be those listed in table III. During environmental test, the units need not be energized or cooled down but shall be subjected to a continuity check for each element.

4.7.1 Temperature shock. The detector-Dewar shall be tested in accordance with MIL-STD-810, Method 503, Procedure I except that the temperature of step 2 shall be -54°C and the temperature of steps 1 and 4 shall be +71°C. Failure to meet requirements of 3.7.1 shall constitute failure of this test.

4.7.2 High temperature. The detector-Dewar shall be tested in accordance with MIL-STD-810, Method 501, Procedure I. The temperature of steps 4 and 5 shall be +71°C. Failure to meet requirements of 3.7.2 shall constitute failure of this test.

4.7.3 Low temperature. The detector-Dewar shall be tested in accordance with MIL-STD-810, Method 502, Procedure I except that the storage temperature of step 2 shall be -62°C and the low operating temperature of step 4 shall be -54°C. Failure to meet requirements of 3.7.3 shall constitute failure of this test.

4.7.4 Shock. The detector-Dewar shall be tested in accordance with MIL-STD-810, Method 516.1, procedure IV, Figure 516.1-1 (100 g's peak value at 11 milliseconds time duration) and Procedure V. Continuity checking during shock testing is not required. Failure to meet requirements of 3.7.4 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

4.7.5 Vibration. The detector-Dewar shall be tested in accordance with MIL-STD-810, Method 514.1, Procedures I and VIII except that the curve shown in figure 1 of this specification shall be used in lieu of the MIL-STD-810 curves. Failure to meet requirements of 3.7.5 shall constitute failure of this test.

4.8 Reliability. An MTBF of 27,600 hours shall be demonstrated (first article inspection only) in accordance with MIL-STD-781, Test Plan IVA, Test level C, when specified in the contract (see 6.2). Failure to demonstrate 27,600 hours MTBF shall constitute failure of this test. Failure shall be defined as any damage (see 6.2) occurring during or at the end of the test. In addition, each detector-Dewar shall successfully complete all tests of table III at the conclusion of this test. Failure of any detector-Dewar to pass any one of the tests in table III shall count as a relevant failure for the purpose of reliability testing.

4.9 Inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine compliance with requirements of Section 5.

4.10 Preconditioning. Each detector-Dewar shall be subjected to preconditioning environments of 3.10 and as augmented by Government-approved test plan to be submitted as part of contract (see 6.2).

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. 2M-1-1-ØØ-00-NV-X-XX-P-ED-A.

5.1.2 Level B. 1Ø-1-1-ØØ-00-NV-X-B7-P-ED-B.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. The detector-Dewar is intended for use in infrared systems.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. First article tests are required except for existing suppliers (see 6.3.9).

(1) Time frame for submission of first article test reports when first article tests are required.

(2) Time frame for approval of first article test reports when first article tests are required.

(3) First article reliability testing if required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

c. Production delivery schedule.

(1) Defined in terms of monthly lots.

(2) Actions required relative to group C failures (see 4.5.2.3.2).

d. Level A or level B preservation and packaging (see section 5).

e. MIL-STD-810B shall be used for environmental tests of 4.7.

f. Requirement for contractor-submitted test plan for preconditioning DI-T-1903 (Part, Component or Subsystem Test Plan(s))(see 4.10).

g. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

h. Deliver all first article samples to - Commander, USAERADCOM, ATTN: DELNV-CM, Fort Belvoir, VA 22060.

i. Group C inspection monthly sample size to be established based on cumulative production totals of concurrent contracts.

j. The average detectivity requirement, 3.6.5.c, is required for procurement of all modules to this specification made directly by the Government. The requirement, 3.6.5.c, may be waived by system contractors ordering modules to this specification for use in the manufacture of completed systems if these systems meet system performance requirements.

6.3 Definitions.

6.3.1 Element. An element is defined as one channel in the 120-element detector-Dewar.

6.3.2 Defective element. A defective element shall be defined as an element which fails to meet any one of the requirements of 3.6.1, 3.6.5, 3.6.6 and 3.6.7. Any elements which are open, shorted, or microphonic are considered defective. Two adjacent elements shorted together are considered to be a single defect.

6.3.3 Strapping. Strapping is defined as electrically connecting a defective element to a nondefective element. The pair of strapped elements thus formed, exhibiting 100 percent crosstalk, shall be considered as a single defect.

6.3.4 Array axis. The array axis is defined as a line on the surface of the array passing through the center of the geometrical areas of each element.

6.3.5 Optical axis. The detector-Dewar's optical axis is defined as a line perpendicular to the plane of the detectors that intersects the array axis at the mid-point of the array.

NOTE: One-half of the active detector elements are located on each side of the optical axis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

6.3.6 Damage. Breakage, loosening, shifting, evidence of corrosion, or failure of any finish, hardware, connection or component; and any degradation of detector-Dewar performance to values less than specified herein.

6.3.7 Nominal detector element size. Defined as 0.0016 by 0.0023 inch.

6.3.8 Specified lot quantity. The specified lot quantity shall be one-month's production unless the contract specifies a different lot basis such as one-week's production or those units processed as a single batch.

6.3.9 Existing suppliers. Suppliers who are currently supplying equipment to this specification with a minimum average monthly rate of 10 units per month over the most current 12-month period. If a supplier cannot maintain the minimum monthly average, a first article test (less reliability) performed annually shall sustain the existing supplier's status when performed on five (5) units manufactured by the supplier explicitly for this test. Upon implementation of this specification, suppliers who are then supplying equipment to specification B2-28A050102, Detector-Dewar DT-594/UA (modified for 120 element application), with the minimum monthly delivery rate specified above shall be considered as existing suppliers.

CUSTODIAN:
Army - ER

PREPARING ACTIVITY:
Army - ER

Project 5855-A221

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

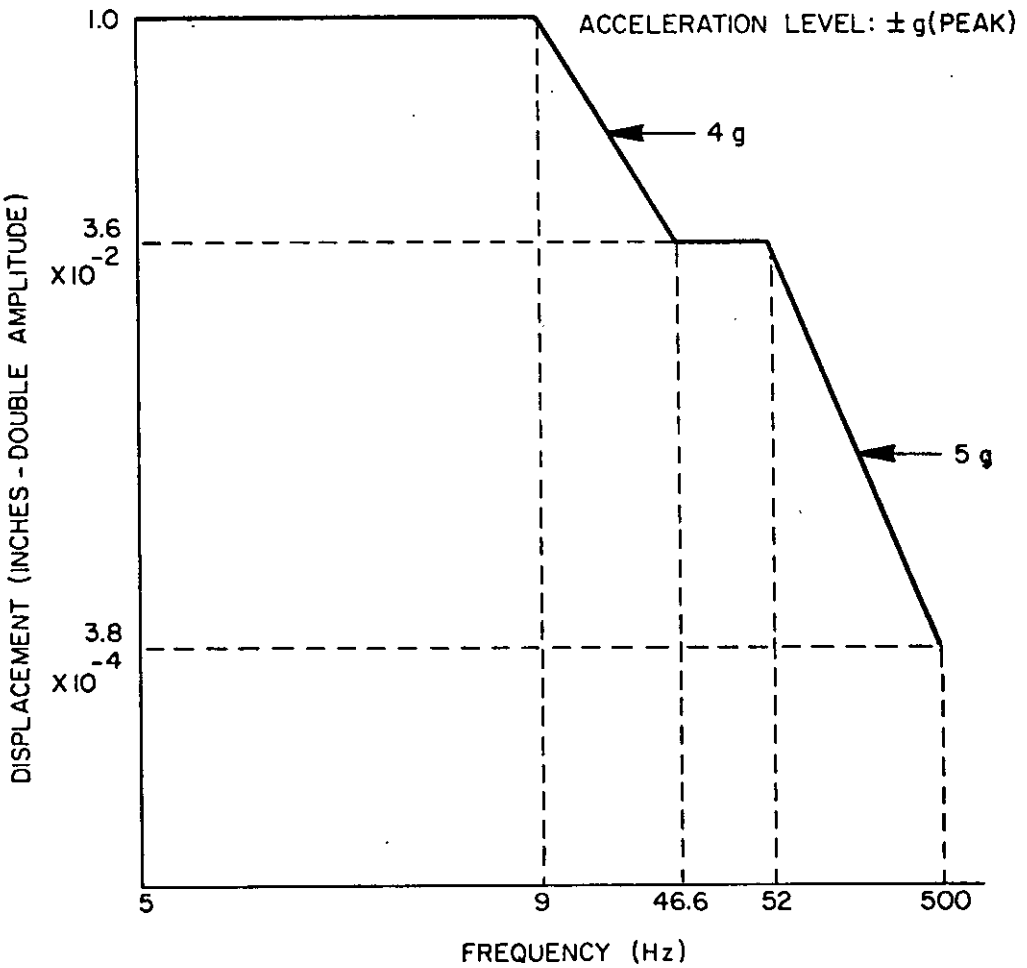


FIGURE 1. VIBRATION TEST PROFILE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND MIL-D-49419

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   [ ] YES [ ] NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:04Z
Check the source to verify that this is the current version before use.